

The Nambu-Schwarz-Ramond (NS-R) formalism:

SuperSpace: $M^{a,b}$ $a: D$, $b: \text{susy}$
 $\{x^\mu, \chi_\mu\}$, $\mu = 0, 1, \dots, D-1$

χ_A^μ : 2D-Grassmann odd

* χ^μ is a D -dim vector and not seen as a space-time spinor

* It is however a 2D worldsheet spinor and $\text{Spin}(2)$ transformations act on its 2 index.

* Supermultiplet: $\{x^\mu, \chi^\mu, N^\mu\}$ N^μ : Scalar Aux. field

$$\int d^2\xi \left(-\frac{1}{2} \partial_\alpha x^\mu \partial^\alpha x_\mu - \frac{i}{2} \bar{\chi}^\mu \partial \chi_\mu + \frac{1}{2} N^\mu N_\mu \right) \eta_{\mu\nu}$$

* I believe we can obtain this action from

$$\int d^4\theta \int d^D x \bar{\Phi} D^\mu D_\mu \Phi$$

* Coupling this action to (Super)gravity:

$$S = \frac{1}{2\pi\alpha'} \int d^2\xi e \left\{ -\frac{1}{2} e^\alpha_a \partial_\alpha x^\mu e^{a\beta} \partial_\beta x^\nu - \frac{i}{2} \bar{\chi}^\mu \partial \chi_\mu + \frac{1}{2} N^\mu N_\mu \right. \\ \left. + \frac{i\kappa}{2} \bar{\psi}_\alpha \partial x^\mu \gamma_\mu \psi^\alpha - \frac{\kappa^2}{16} \bar{\psi}_\alpha \gamma^\mu \psi^\alpha \bar{\chi}^\mu \chi_\mu \right\}$$

$$\partial \equiv \partial^\alpha e_\alpha^a \partial_a$$

* See West Notes chap 21 for derivation.

* This action has the following symmetries:

(a) Diffeomorphism invariance:

$$\delta e_\alpha^a = f^\beta \partial_\beta e_\alpha^a + \partial_\alpha f^\beta e_\beta^a, \quad \delta \psi_{\alpha A} = f^\beta \partial_\beta \psi_{\alpha A} + \partial_\alpha f^\beta \psi_{\beta A}$$

$$\delta x^\mu = f^\beta \partial_\beta x^\mu, \quad \delta \chi_A^\mu = f^\beta \partial_\beta \chi_A^\mu$$

$$\delta N^\mu = f^\beta \partial_\beta N^\mu$$

$f^\alpha = f^\alpha(\xi)$: local worldsheet dependent, parameter.

(b) Local two-dim Lorentz invariance:

$$\delta e_\alpha^a = -\omega_b^a e_\alpha^b, \quad \delta \chi_A^M = \frac{1}{4} (\omega^{ab} \gamma_{ab})_A^B \chi_B^M$$

$$\delta \psi_A = \frac{1}{4} (\omega^{ab} \gamma_{ab})_A^B \psi_B$$

$\rightarrow \chi^m$ and N^m are inert

(c) Local 2D SUSY:

$$\delta \chi^m = i \bar{\epsilon} \chi^m, \quad \delta \chi^m = (\tilde{D} \chi^m + N^m) \epsilon, \quad \delta N^m = i \bar{\epsilon} \tilde{D} \chi^m$$

$$\delta e_\alpha^a = i \kappa \bar{\epsilon} \gamma^a \psi_\alpha, \quad \delta \psi_\alpha = \frac{2}{\kappa} D_\alpha \epsilon$$

where

$$\hat{D}_\alpha \chi^m = \partial_\alpha \chi^m - \frac{i\kappa}{2} \bar{\psi}_\alpha \chi^m$$

$$\hat{D}_\beta \chi^m = (D_\beta \chi^m - \frac{\kappa}{2} \hat{D} \chi^m + N^m) \psi_\beta \quad \left| \quad D_\beta \epsilon = (\partial_\beta + \frac{1}{4} \omega_{\beta ab} \gamma^{ab}) \epsilon \right.$$

$$D_\beta \chi^m = (\partial_\beta + \frac{1}{4} \omega_{\beta ab} \gamma^{ab}) \chi^m$$

(d) Weyl symmetry:

$$\delta e_\alpha^a = \Lambda e_\alpha^a, \quad \delta \psi_\alpha = \frac{1}{2} \Lambda \psi_\alpha$$

$$\delta \chi^m = 0, \quad \delta \chi^m = -\frac{1}{2} \Lambda \chi^m, \quad \delta N^m = -\Lambda N^m$$

(e) Local S-symmetry:

$$\delta \psi_\alpha = \gamma_\alpha \zeta$$

where ζ_A is a Majorana spinor which depends on ζ^α in an arbitrary way.

(f) Poincaré group: (locally)

$$\delta \chi^m = -\Lambda_\nu^m \chi^\nu, \quad \delta \chi^m = -\Lambda_\nu^m \chi^\nu \dots$$

$\rightarrow$ This is the only rigid symmetry.

* We can now look at the Action and obtain E.O.M:

χ^m E.O.M:

$$\frac{\delta \mathcal{L}}{\delta (\partial_\sigma \chi^\sigma)} = (-e \partial_\alpha \chi^\sigma e^{\alpha\gamma} + \frac{i\kappa}{2} \bar{\psi}_\alpha \gamma^\gamma \gamma^\sigma \chi^\sigma) \eta_{0\sigma}$$

$$\partial_\gamma \left(\frac{\delta \mathcal{L}}{\delta (\partial_\sigma \chi^\sigma)} \right) = \partial_\gamma (-e \partial_\alpha \chi_\sigma e^{\alpha\gamma}) + \frac{i\kappa}{2} \partial_\gamma (\bar{\psi}_\alpha \gamma^\gamma \gamma^\sigma \chi_\sigma e) = 0$$

$$\partial_\gamma (\partial_\alpha \chi_\sigma e^{\alpha\gamma}) - \frac{i\kappa}{2} \partial_\beta (e \bar{\psi}_\alpha \gamma^\gamma \gamma^\sigma \chi_\sigma) = 0$$

$$\chi^m \text{ E.O.M.: } \frac{\delta \mathcal{L}}{\delta \partial_\gamma \chi^\sigma} = -e^{\frac{i}{2}} \bar{\chi}^m \gamma^\sigma e_a^\gamma \eta_{m\sigma}$$

$$\partial_\gamma \left(\frac{\delta \mathcal{L}}{\delta \partial_\gamma \chi^\sigma} \right) = -\partial_\gamma \left(e^{\frac{i}{2}} \bar{\chi}^m \gamma^\sigma e_a^\gamma \eta_{m\sigma} \right)$$

$$\delta \mathcal{L} / \delta \chi^\sigma = \frac{i\kappa}{2} \bar{\psi}_a \not{\partial} \chi^m \gamma^\sigma \eta_{m\sigma} - \frac{\kappa^2}{16} \bar{\psi}_a \gamma^\beta \gamma^\alpha \psi_\beta \bar{\chi}^m \eta_{m\sigma}$$

$$\frac{i}{2} \partial_\gamma (e \bar{\chi}^m \gamma^\sigma e_a^\gamma \eta_{m\sigma}) - \frac{i\kappa}{2} \bar{\psi}_a \not{\partial} \chi^m \gamma^\sigma \eta_{m\sigma} + \frac{\kappa^2}{16} \bar{\psi}_a \gamma^\beta \gamma^\alpha \psi_\beta \bar{\chi}^m \eta_{m\sigma} = 0$$

* To Obtain χ^m E.O.M in the text ~~use~~ ^{use}

$$\frac{i\kappa}{2} \bar{\psi}_a \not{\partial} \chi^m \gamma^\sigma \eta_{m\sigma} = \frac{i\kappa}{2} \bar{\chi}^m \gamma^\sigma \not{\partial} \chi^m \psi_a$$

$$\text{and so } \delta \mathcal{L} / \delta (\partial_\gamma \bar{\chi}^m) = 0$$

$$\frac{\delta \mathcal{L}}{\delta \bar{\chi}^\sigma} \Rightarrow \not{\partial} \chi^m - \frac{\kappa}{2} \gamma^\alpha \not{\partial} \chi^m \psi_\alpha - i \frac{\kappa^2}{2} \bar{\chi}^m \bar{\psi}_a \gamma^\beta \gamma^\alpha \psi_\beta = 0$$